

Siddartha Reddy Munugala

FULL-STACK ENGINEER • AI PRODUCT BUILDER • FOUNDER
Vermillion, SD • Siddarth.lbf341@gmail.com • github.com/Redsid5 • [LinkedIn](#) • vibein.us • arenahq.us

PROFILE

CS student who builds and ships. Two live products, 1,621 users, and a track record of owning the full stack — architecture, debugging, deployment, and iteration — from zero. Currently building EKO, a proprietary AI dubbing and localization platform, and PRISM, an ML system for predicting post-merge engineering risk. Looking for an early-stage environment where shipping speed and technical ownership matter.

EDUCATION

University of South Dakota Vermillion, SD • Expected May 2027
B.S. Computer Science — Minor in Innovation & Entrepreneurship

STACK

Languages: Python, TypeScript, JavaScript, SQL • **Frameworks:** Next.js 14, React, Node.js, Tailwind CSS
AI / ML: XGBoost, SHAP, PyTorch, Claude API, Gemini API, OpenAI API, LangChain, Whisper • **Audio:** FFmpeg, speech alignment, waveform processing
Infra: Supabase, PostgreSQL, Redis, Docker, Vercel, Git, GitHub Actions • **Protocols:** WebRTC, WebSockets, REST, Supabase Realtime

SHIPPED PRODUCTS

EKO — AI Dubbing & Localization Platform 2026 – Present
Founder • Full-Stack + AI/Audio Pipeline

- > Building a proprietary audio intelligence pipeline — transcription, dialogue-aware translation, voice preservation, emotion matching, and timing alignment — with no dependency on third-party dubbing APIs.
- > Architected queue-based job processing for parallel audio segmentation, synthesis, and multilingual output orchestration; frontend in Next.js with timeline review and export tooling.
- > Stack: Python, FFmpeg, Whisper-style transcription, Next.js, TypeScript, cloud storage, and async job queues.

VibeIn — Anonymous Professional Network Jan 2025 – Present
Founder • Next.js 14 / Supabase / Twilio / Gemini AI • vibein.us

- > Shipped an anonymous matchmaking platform built on peer recommendation matrices and mutual intent signals; grew to 821 registered users pre-launch.
- > Resolved a production cascade of 8+ critical bugs — RLS auth failures, Gemini API deprecation, broken SMS/WhatsApp notifications, silent Realtime failures — in a single sprint ahead of a scheduled user test.
- > Implemented row-level security, real-time chat, and Supabase Realtime; maintained 99%+ uptime on Vercel.

ArenaHQ — Live Competitive Coding Platform Jan 2025 – Present
Founder • Next.js / Supabase WebSockets / WebRTC / CodeMirror 6 • arenahq.us

- > Built a live 1v1 coding tournament engine with real-time code evaluation, P2P camera authentication for live proctoring, and a recruiter fast-track portal bypassing traditional screening loops.
- > Grew to 800 registered users pre-beta; 20% active retention across combined VibeIn + ArenaHQ registry of 1,621 users.

RESEARCH & TOOLS

PRISM — PR Risk Intelligence & Structural Modeling 2026
> ML pipeline predicting post-merge engineering risk from 45 engineered features (AST complexity via tree-sitter, dependency graphs via NetworkX); XGBoost + SHAP explainability; validated at $AUC \geq 0.80$, $\Delta AUC < 0.05$ cross-repo.
> Authored two formal research proposals with mathematical appendices — USD faculty review and external mentorship context.

COGN — Cognitive Integrity Layer for AI Codebases 2026
> Open-source CI/CD GitHub Action scoring PRs for "cognitive drift" — measurable reasoning shifts introduced by AI-generated code — against repo baselines using three custom primitives.

ALASKA — Encrypted AI Assistant 2026 • Top-10 Hackathon
> 48-hour build: AES-256-GCM encrypted assistant for military veterans with VA routing, crisis detection, and accessibility modes.

OTHER EXPERIENCE

Production Operations Staff — USD Athletics Aug 2025 – Present
> Operate live-event audio, stadium displays, and broadcast cues for NCAA events under zero-tolerance real-time constraints. Part-time on-campus.

IT Coordinator & Publicity Chair — NSLS Executive Board Sep 2024 – Present
> Built member tracking pipelines, recruitment dashboards, and chapter admin portal; authored 10-week strategic operational plan for USD INFORMS.